Problem Set 1: Input and Output in Python

For this assignment, you will use Python and Visual Studio Code (VSC) to create a computer program. Submit a Python program script file with the solutions (file extension is py) in Blackboard by Sunday at 11:59 PM. The objective of this coursework is to establish your Python working environment and begin familiarizing yourself with working in Python. This work utilizes some basic code provided in the course by demonstrating the ability to display information, collect input, and store information in objects.

Installing Python and VSC

Throughout the course, you will work with Python using the integrated development environment VSC. Use the videos provided as learning materials to install and prepare the necessary software for this course. There are additional videos that may help you understand how to interact with the VSC software.

Establish the Working Directory

It is necessary to establish a working directory for Python files in VSC. In VSC, go to preferences, then settings, and use the dropdown search bar to look for 'Execute in File Dir.' The search will prompt you with Python and a checkbox next to 'Execute in File Dir,' if it is not checked, check the box. In this course, you will use external files and import them into your script file. When you have external files, save them to the same folder as the script file.

After completing this task, create a new script file named **`working_directory_check.py`**. This file will be used to validate your current working directory. You will not submit this file. Enter the information shown in Figure 1 in this file. Update the block comments where directed. (Three quotation marks surround block comments). After you have saved the code, execute this file. The output in the terminal should give you the

Figure 1

Content For Validation of the Working Directory

```
"""
File created to validate the working directory.
Executing this script file will print the location of
this script file, if VSC was setup as directed.

Enter the date this file was made here.
Enter your name, as the author of this script here.
"""

import os # module for OS interaction

print("\nThe current directory is", os.getcwd(), "\n")
```

Note. This figure contains the exact content necessary to test the update made to the VSC environment.

path to the directory where **`working_directory_check.py`** is saved. If the terminal output does not provide the expected path, redo the tasks in this section.

Script Requirements

When writing and saving your programming files, ensure that you heed the instructions. For all programming coursework, specific things must be placed in the programming code. The beginning of the script file must include the assignment name, the purpose of the programming code, the date it was written or revised, and who wrote the code. These comments can be written as inline comments or block comments. Figure 2 depicts an example of what this might look like in your file. When saving your program, name the file **`assignment_1.py`**. After completing and testing your script file, submit **`assignment_1.py`** in Blackboard.

Figure 2

Content Needed in Every Script File

```
"""
Assignment 1: this is a basic input and output file
September 22, 2022
John Smith
"""
```

Note. This contains the name of the assignment (Assignment 1), purpose of this assignment (this is a basic input and output script), date (September 22, 2022), and the name of the developer (John Smith).

Interacting with a User

All programs, like data analysis, begin with a problem to solve, a question to answer, or something to prove (you might call this a hypothesis). This program is no exception. The following information specifies the requirements of this program, along with an example of what the output might look like in the console and some hints for completing this work. You have been tasked with creating a solution to collect specific information from program users. There are three required objectives to solve this problem. Ask the user to enter their name. Ask the user whether they know the programming language Python. The last requirement is to print their name to the terminal when you thank them for participating. An example of what you might see in the console when your script runs is shown in Figure 3. As long as you collect the necessary input and present the necessary output, feel free to be creative. Should you run into problems determining what commands you need to use, check out the input and output section of the Python Programming wiki. To reuse information in a program, you have to store that information. You may call this an object, a variable, or a name. If nothing else, you may find the variables and strings section of the Python Programming wiki helpful.

Figure 3

Console Output for Expected Script Submission

```
What is your name?
Kathy
Do you know Python?
sure
Thanks for answering my questions, Kathy.
```

Note. This is an example of what the expected output will look like when you run the program and a user answers the questions. The text is yellow reflects the output from the program. The text in white reflects the user's input.